Discrete Random Variables

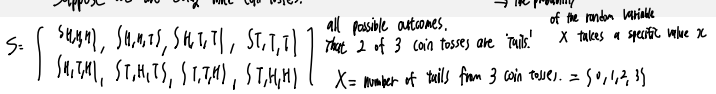
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LESE152

Discrete Random Variables: Several Concepts

- Random Variables
- The PMF and the CDF
- Expected Value
- Variance and Standard Deviation
- Standard Discrete Distributions
- Practice Questions

Random Variables



- Random variable (X): A number whose value depends on a random outcome. Formally, it is a function that assigns a number to each outcome in the sample space.
- Economics is a mostly quantitative field: prices, quantities, interest rates, unemployment rates, and GDP are all numbers. When a random outcome is described by a number, we call it a random variable.
- Notation: A capital letter X denotes the random variable, and a lowercase letter x denotes a particular value it can take.
- Examples: The number of heads in 10 coin flips, a farm's harvest, next month's sales, or an indicator equal to 1 if a loan defaults and 0 otherwise.

Random Variables in Roulette

Recall the roulette wheel from Lecture 4: the outcome is the slot $b \in \{0, 1, \ldots, 36\}$. We can define several random variables from the same outcome.

- The number the ball lands on: b.
- An indicator for whether a bet on red wins:

$$r = \begin{cases} 1 & \text{if } b \in Red \\ 0 & \text{otherwise} \end{cases}$$

- The net profit from a \$1 bet on red, y_{Red}: $+1$ if the ball lands on red and -1 otherwise.
- The net profit from a \$1 bet on 14, y_{14}: $+35$ if the ball lands on 14 and -1 otherwise.

All four are functions of the same outcome b, so they are related to each other.

Discrete and Continuous Random Variables

- Support: The set of values that the random variable can actually take, i.e., the values with positive probability.
- Roulette: The support of b is $\{0, 1, \ldots, 36\}$, the support of r is $\{0, 1\}$, and the support of y_{14} is $\{-1, 35\}$.
- Discrete random variable: The possible values can be counted or listed, with gaps in between. Examples: the number of customers arriving in an hour, the number of defective items in a batch.
- Continuous random variable: The possible values fill an interval, so they cannot be listed. Examples: a price, a wage, the amount of rainfall. We study these in Lecture 6.

Probability Mass Function (PMF)

- Probability mass function: The probability that X takes each value,

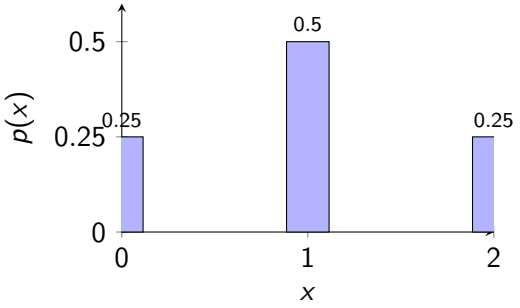
$$p(x) = P(X = x).$$

- It can be written as a table, a graph, or a formula. For a valid PMF:
  1. $0 \leq p(x) \leq 1$ for every x.
  2. $\sum_x p(x) = 1$, where the sum is over the support.
- The probability of any event is the sum of the PMF over the values in that event:

$$P(X \in A) = \sum_{x \in A} p(x).$$

This is the same rule as in Lecture 4: add up the probabilities of the elementary events.

PMF: Two Coin Tosses



Toss a fair coin twice. The sample space is $\{HH, HT, TH, TT\}$. Let $X =$ the number of heads.

x	0	1	2
$p(x)$	0.25	0.50	0.25

$p(1) = P(HT) + P(TH) = 0.5.$

What is $P(X \geq 1)$? $0.50 + 0.25 = 0.75.$

PMF: Roulette

Assume a fair wheel, so each slot has probability $1/37$.

- The PMF of b: $p_b(x) = 1/37$ for each $x \in \{0, 1, \ldots, 36\}$.
- The PMF of r:

$$p_r(1) = P(Red) = 18/37 \approx 0.486, \qquad p_r(0) = 19/37 \approx 0.514.$$

- The PMF of y_{14}:

$$p_{14}(35) = P(b = 14) = 1/37 \approx 0.027, \qquad p_{14}(-1) = 36/37 \approx 0.973.$$

- Mode: The most likely value. The mode of r is 0, and the mode of y_{14} is -1: the most likely result of either bet is losing.

Cumulative Distribution Function (CDF)

- Cumulative distribution function: The probability that X is at most x,

$$F(x) = P(X \leq x) = \sum_{s \leq x} p(s).$$

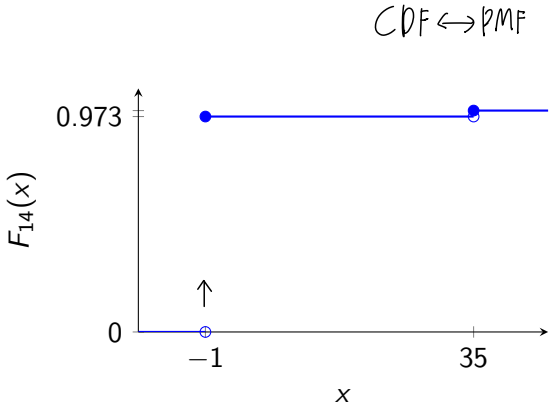
- For a discrete random variable, the CDF is a step function. It jumps up by $p(x)$ at each value in the support and is flat in between.
- Properties: $0 \leq F(x) \leq 1$, F never decreases, and F runs from 0 (far left) to 1 (far right).
- Two coin tosses:

	$250	$500	$1,000
$p(X=x)$	0.3	0.5	0.2
$F(X=x)$	0.3	0.8	1

$F(X=m)=1$

x	0	1	2
$p(x)$	0.25	0.50	0.25
$F(x)$	0.25	0.75	1

CDF: The Bet on 14



The support of y_{14} is $\{-1, 35\}$, so

$$F_{14}(x) = \begin{cases} 0 & x < -1 \\ 36/37 \approx 0.973 & -1 \le x < 35 \\ 1 & x \ge 35 \end{cases}$$

The CDF jumps at each point in the support, and the size of each jump equals the PMF at that point.

Interval Probabilities

$P(\$250 < X \le \$1{,}000) = F(\$1{,}000) - F(\$250)$

$\neq P(\$250 < X < \$1{,}000)$ $= 1 - 0.3 = 0.7$

$F(x=2) = P(X \le 2)$

- The CDF gives the probability of any interval:

$$P(a < X \le b) = F(b) - F(a).$$

- Be careful with $<$ and $\le$: a discrete random variable can equal the endpoint exactly. Adjust with the PMF, e.g.,

$$P(a \le X \le b) = F(b) - F(a) + p(a).$$

- Roulette: $F_b(1) = 2/37$ and $F_b(36) = 1$, so

$$P(1 < b \le 36) = 1 - 2/37 = 35/37, \qquad P(1 \le b \le 36) = 35/37 + 1/37 = 36/37.$$

- Complement rule: $P(X > a) = 1 - F(a)$.

Expected Value

- Expected value (mean): The probability-weighted average of the possible values,

$$\mu = E(X) = \sum_x x\, p(x).$$

- Multiply each value by its probability, then add everything up.
- Interpretations:
  - A weighted average: each value is weighted by how likely it is.
  - A measure of central tendency: a typical value of X.
  - A long-run average: the average over many independent repetitions.

Expected Value: Examples

- Daily sales X (in units) have the following PMF:

x	10	20	30
$p(x)$	0.3	0.5	0.2

$$E(X) = 10(0.3) + 20(0.5) + 30(0.2) = 3 + 10 + 6 = 19 \text{ units.}$$

  Note that 19 is not a possible value of X. The mean does not have to be in the support.
- Roulette, bet on red: $E(r) = 0 \times 19/37 + 1 \times 18/37 = 18/37 \approx 0.486$.
- Roulette, number: $E(b) = (0 + 1 + \cdots + 36)/37 = 18$.

The House Advantage

→ Expected value

X = Revenue from Crop harvest

	High P. $4	Low P. $1
Good H. 100kg	0.1	0.3
Bad H. 50kg	0.4	0.2

1) support $\{50, 100, 200, 400\}$

2) $E(X) = ?$ $E(X) = 10 + 30 + 80 + 40 = 160$

3) $E(X \mid \text{Good H.}) = ?$

- Bet \$1 on 14:

$$E(y_{14}) = (-1) \times \frac{36}{37} + 35 \times \frac{1}{37} = -\frac{1}{37} \approx -0.027.$$

$P(H,P \mid G,4) = \frac{P(H,P \cap G,H)}{P(G,H)} = \frac{0.1}{0.4}$

$P(L,P \mid G,H) = \frac{0.3}{0.4} = \frac{3}{4}$

$E(X \mid G,H) = \$100 \cdot P(L,P \mid G,H) + \$400 \cdot P(H,P \mid G,H)$

$95 + 100$

- Bet \$1 on red:

$$E(y_{Red}) = (-1) \times \frac{19}{37} + 1 \times \frac{18}{37} = -\frac{1}{37} \approx -0.027.$$

- Both bets lose about 2.7 cents per dollar on average. This is the house advantage: over many bets, the casino earns about 2.7% of all money wagered.
- The two bets have the same mean, but they are clearly not the same bet. We need a measure of spread (next section).

$E(X \mid \text{Good H.})$

⇒ Conditional Expected value of the return given that the harvest is 'Good'

Functions of a Random Variable

$Y = X + 1$ ← Random Variable

Now let's define a new random variable Y, Revenue of farming 100 acres $\Rightarrow Y = 100 \times X + 20$

$E(Y) = (100 \times \$50 \times P(X=50) + (100 \times \$100 \times P(X=100) + \cdots$ (+20) (+20)

- Any function of a random variable is also a random variable. If X is random, so are X^2, $\ln X$, and $Y = a + bX$.
- Expected value of a function: Use the PMF of X,

$$E[g(X)] = \sum_x g(x)\, p(x).$$

- Example: $E(y_{14}^2) = (-1)^2 \times \frac{36}{37} + 35^2 \times \frac{1}{37} = \frac{1261}{37} \approx 34.08$.
- Linear function: $Y = a + bX$ for constants a and b. Otherwise, it is a nonlinear function.
- Economic models are full of functions of random variables: revenue, profit, and utility all depend on random quantities.

Properties of Expected Values

$E(Y) = 100 \cdot \$50 \cdot P(X=50) + 100 \cdot \$100 \cdot P(X=100) + \cdots$

$= 100\,(\$50 \cdot P(X=50) + \$100 \cdot P(X=100) + \cdots)$

Suppose

$P = 3 \cdot X + 4 \cdot Y + 100$

$= 3X + 4Y + 100$

X = profit from growing corn ($/acre)

Y = profit from growing soybean ($/acre)

$E(X) = \$200/\text{acre}$

$E(Y) = \$150/\text{acre}$

If a farm grows 3 acres of corn, 4 acres of bean, expected profit, this from Government

For constants a, b and random variables X, Y:

- $E(a) = a$.
- Linearity: $E(a + bX) = a + b\,E(X)$. We only need $E(X)$, not the whole distribution.
- $E(X + Y) = E(X) + E(Y)$. This holds whether or not X and Y are independent.
- Roulette: $y_{Red} = 2r - 1$, so $E(y_{Red}) = 2E(r) - 1 = 2(18/37) - 1 = -1/37$.
- Never take the expected value inside a nonlinear function: in general $E[g(X)] \neq g(E(X))$.
  - $E(X^2) \neq [E(X)]^2$, $\quad E(1/X) \neq 1/E(X)$, $\quad E(\ln X) \neq \ln E(X)$.
  - Roulette: $E(b) = 18$ is red, but that does not mean a bet on red is expected to win.

$E(P) = 3E(X) + 4E(Y) + 100$

$= 600 + 600 + 100$

$= 1300$

Expected Value and Decisions

$E(a) = 500 \times 1 = 500$ $E(b) = 0 \cdot 0.5 + 1000 \cdot 0.5 = 500$

- Choose: (a) 500 for sure, or (b) a coin flip for 0 or 1,000?
- Both have the same expected value, but most people choose (a). They are risk-averse.
- Expected utility: People weigh outcomes by a utility function $u(x)$,

$$E[u(X)] = \sum_x u(x)\, p(x).$$

  A concave u penalizes spread. Since u is nonlinear, $E[u(X)] \neq u(E(X))$.
- St. Petersburg game: Win 2^k if the first head appears on flip k. The expected value is infinite, yet few people would pay much to play. Bernoulli's resolution (1738) founded expected utility theory.

Variance and Standard Deviation

- Variance: The expected squared deviation from the mean,

$$\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = \sum_x (x-\mu)^2\, p(x).$$

- Shortcut formula (often easier):

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \sum_x x^2\, p(x) - \mu^2.$$

- Standard deviation: $\sigma = \sqrt{\mathrm{Var}(X)}$. It is measured in the same units as X, which makes it easier to interpret.
- The variance is always non-negative. Low variance means similar results each time; high variance means results vary a lot. In economics and finance, the variance of returns is the standard measure of risk.

$$\sigma^2 = E\left[(X-\mu)^2\right] \rightarrow \sum_x (x-\mu)^2 \cdot p(x)$$

($E(x)$)

Squared deviation from mean

$E(g(x))$ $\llcorner (x-\mu)^2$		HP	LP
	GH	0.1	0.4
	BH	0.3	0.2

Suppose = {\$50, \$100, \$150, \$200}

$E(x) = \$50 \cdot 0.2 + \$100 \cdot 0.4 + \cdots + \$400 \cdot 0.1$

$= \$110$

$Var(x) = E(X-110)^2$

$(\$50 - 150)^2 \cdot P(X=50)$

$+ (\$100 - 150)^2 \cdot P(X=100) \cdots$

$+ (\$200 - 150)^2 \cdot P(X=150)$

$+ (\$400 - 150)^2 \cdot P(X=200)$

$= \$100^2 \cdot 0.2 + \$60^2 \cdot 0.4 + \$50^2 \cdot 0.3 + \$250^2 \cdot 0.1$

$= 10,000$

$\sigma = \$100$

$Var(X) = E(X^2) - [E(X)]^2$

$= \$32,500 - (\$150)^2$

$22,500$

$= \$10,000$

$E(X) = \$150$

$E(X^2) = \$50^2 \cdot 0.2 = 500$

$+ \$100^2 \cdot 0.4 = 4000$

$+ \$200^2 \cdot 0.3 = 12000$

$+ \$400^2 \cdot 0.1 = 16000$

$= 32500$

Variance: An Example

A coffee shop sells X specialty drinks per hour, with the following PMF.

x	$p(x)$	$x\,p(x)$	$(x-\mu)^2\,p(x)$	$x^2\,p(x)$
10	0.1	1	$100 \times 0.1 = 10$	10
15	0.2	3	$25 \times 0.2 = 5$	45
20	0.4	8	$0 \times 0.4 = 0$	160
25	0.2	5	$25 \times 0.2 = 5$	125
30	0.1	3	$100 \times 0.1 = 10$	90
Sum	1	$\mu = 20$	$\sigma^2 = 30$	430

- Definition: $\sigma^2 = 10 + 5 + 0 + 5 + 10 = 30$.
- Shortcut: $\sigma^2 = 430 - 20^2 = 30$. Either way, $\sigma = \sqrt{30} \approx 5.48$ drinks.

Red or 14? Same Mean, Different Risk

- r is 0 or 1: $\mathrm{Var}(r) = (0 - 0.486)^2(0.514) + (1 - 0.486)^2(0.486) \approx 0.25$.
- y_{14}: Using the shortcut and $E(y_{14}^2) \approx 34.08$,

$$\mathrm{Var}(y_{14}) = 34.08 - (-0.027)^2 \approx 34.08, \qquad \sigma_{14} \approx 5.84.$$

- $y_{Red} = 2r - 1$: $\mathrm{Var}(y_{Red}) = 4 \times 0.25 \approx 1.0$, $\sigma_{Red} \approx 1.0$ (rule on the next slide).
- Both bets lose 2.7 cents per dollar on average, but:
  - Players who bet on red have similar results: most lose their money slowly.
  - Players who bet on 14 have very different results: a few win big, most lose quickly.

Variance of a Linear Function

$\mathrm{Var}(a+bX) = b^2\mathrm{Var}(X)$

Y

$\mathrm{Var}(Y) = E(Y-E(Y))^2$

$= E((a+bX) - (a+b(E(X))$

$= E((bX - bE(X))^2) = E[(b(X-E(X))^2] = E[b^2 \cdot (X-E(X))^2]$

- For constants a and b:

$$\mathrm{Var}(a+bX) = b^2\,\mathrm{Var}(X), \qquad \mathrm{sd}(a+bX) = |b|\,\mathrm{sd}(X).$$

- Adding a constant shifts the distribution but does not change its spread. Multiplying by b scales the variance by b^2. Also, $\mathrm{Var}(a) = 0$.
- Daily sales from before: $E(X) = 19$, $\mathrm{Var}(X) = 49$, $\sigma = 7$. Suppose each unit earns 5 and the fixed cost is 40, so profit is $Y = 5X - 40$.

$$E(Y) = 5(19) - 40 = 55, \qquad \mathrm{Var}(Y) = 5^2 \times 49 = 1225, \qquad \sigma_Y = 35.$$

- The fixed cost shifts expected profit but does not change risk.

Standardization

- Standardizing: Subtract the mean and divide by the standard deviation,

$$Z = \frac{X - \mu}{\sigma}.$$

- Z is a linear function of X, so by the rules above,

$$E(Z) = 0, \qquad \mathrm{Var}(Z) = \mathrm{sd}(Z) = 1.$$

- Z measures how many standard deviations X is from its mean. $Z = 0.5$: half a standard deviation above the mean. $Z = -3$: three standard deviations below.
- Daily sales: a day with 30 units has $Z = (30 - 19)/7 \approx 1.57$.
- We will use standardization heavily with the normal distribution.

Insurance: An Example

An insurer covers a loss of 10,000 that occurs with probability 0.02. Let X be the payout.

- $E(X) = 10000 \times 0.02 = 200$. This is the actuarially fair premium.
- $E(X^2) = 10000^2 \times 0.02 = 2{,}000{,}000$, so

$$\mathrm{Var}(X) = 2{,}000{,}000 - 200^2 = 1{,}960{,}000, \qquad \sigma = 1400.$$

- The standard deviation is seven times the mean: a policy is a very risky asset for the insurer to hold on its own.
- The premium actually charged exceeds 200 to pay for risk-bearing and administration. Customers pay it anyway because they are risk-averse.

Standard Distributions

- Some distributions appear so often in applications that they have names. We write

$$X \sim \text{DistributionName(parameters)},$$

  where $\sim$ reads "has the distribution" and the parameters are numbers that pin down the exact PMF.
- Each named distribution is a family: a different PMF for each value of the parameters.
- We derive the mean and variance once, and then use the results every time we recognize the distribution.
- Today: Bernoulli, binomial, Poisson, and discrete uniform.

Bernoulli Distribution

- $X \sim \text{Bernoulli}(p)$: $X = 1$ (success) with probability p and $X = 0$ (failure) with probability $1 - p$.
- Mean and variance:

$$E(X) = 0(1-p) + 1(p) = p, \qquad \text{Var}(X) = E(X^2) - p^2 = p - p^2 = p(1-p).$$

- Any indicator of an event A is Bernoulli with $p = P(A)$. Roulette: $r \sim \text{Bernoulli}(18/37)$.
- The workhorse for binary outcomes: employed or not, default or not, buy or not.
- The variance $p(1-p)$ is largest at $p = 0.5$, where uncertainty is greatest.

Binomial Distribution

- A binomial experiment has four features:
  1. A fixed number of trials, n.
  2. Each trial has two outcomes, success or failure.
  3. The probability of success, p, is the same on every trial.
  4. The trials are independent.
- $X \sim \mathrm{Binomial}(n, p)$: the number of successes in the n trials,

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n, \qquad \binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

- Why: each sequence with x successes has probability $p^x(1-p)^{n-x}$ by independence, and there are $\binom{n}{x}$ such sequences.
- Mean and variance: $E(X) = np, \quad \mathrm{Var}(X) = np(1-p)$.

Binomial: Quality Control

A factory produces light bulbs with a 5% defect rate. In a sample of 20 bulbs, let $X =$ the number of defective bulbs, so $X \sim \text{Binomial}(20, 0.05)$.

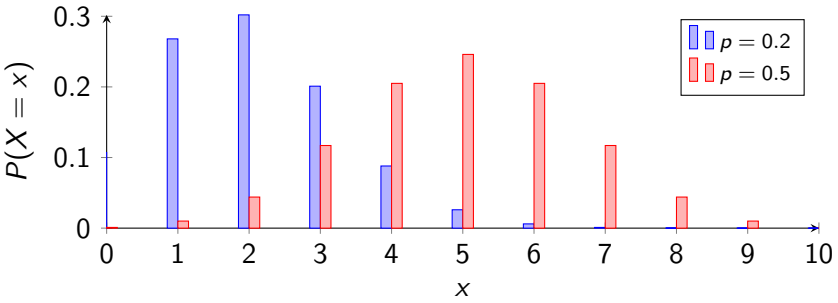
- Exactly 2 defective:

$$P(X = 2) = \binom{20}{2}(0.05)^2(0.95)^{18} = 190 \times 0.0025 \times 0.3972 \approx 0.189.$$

- Expected number of defectives: $E(X) = 20 \times 0.05 = 1$.
- Standard deviation: $\sigma = \sqrt{20 \times 0.05 \times 0.95} \approx 0.975$.
- At least one defective: use the complement rule,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.95^{20} \approx 0.64.$$

Binomial Shapes



- With $n = 10$, the binomial is right-skewed for $p = 0.2$ and symmetric for $p = 0.5$.
- As n grows, every binomial becomes bell-shaped. This previews the normal approximation and the Central Limit Theorem.

When the Binomial Fails

- The binomial requires independent trials with the same p. Check both before using it.
- Sampling without replacement from a small population changes p from draw to draw (recall the M&Ms in Lecture 4). The hypergeometric distribution handles this case.
- Loan portfolio: 100 loans, each defaulting with $p = 0.05$. If defaults were independent, $\sigma = \sqrt{100 \times 0.05 \times 0.95} \approx 2.2$, and 20 or more defaults would be essentially impossible.
- But defaults share common shocks such as recessions, so they are not independent. The number of defaults is then far more variable, and 20 or more becomes plausible.
- Assuming independence when it fails understates risk. This was a central lesson of the 2008 financial crisis.

Poisson Distribution

- $X \sim \text{Poisson}(\lambda)$: the number of events in a fixed interval of time or space, when events occur independently at a constant average rate λ,

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

- Mean and variance: $E(X) = \text{Var}(X) = \lambda$. The mean equals the variance.
- Examples: customers arriving per hour, insurance claims per day, traffic accidents per week, defects per square meter.
- The rate must match the interval. If there are 2 accidents per week on average, the number in two weeks is Poisson with $\lambda = 4$.
- The Poisson approximates the binomial when n is large and p is small, with $\lambda = np$.

Poisson: Examples

- A bank sees an average of 3 customers every 15 minutes at lunch. $X \sim \text{Poisson}(3)$.

$$P(X = 2) = \frac{e^{-3}3^2}{2!} \approx 0.224, \qquad P(X \leq 2) = e^{-3}\left(1 + 3 + \frac{9}{2}\right) \approx 0.423.$$

  The standard deviation is $\sqrt{3} \approx 1.73$ customers.
- A call center receives an average of 4 complaints per day. $X \sim \text{Poisson}(4)$.
  - $P(\text{no complaints}) = e^{-4} \approx 0.018$.
  - Staffing: $P(X > 8) = 1 - P(X \leq 8) \approx 0.021$. A team sized for 8 complaints is overwhelmed on about 2% of days.
- Excel: `POISSON.DIST(x,λ,FALSE)`. R: `dpois(x,lambda)` and `ppois(x,lambda)`.

Discrete Uniform Distribution

- Discrete uniform: Puts equal probability on every value in a finite set S,

$$p(x) = \frac{1}{|S|} \quad \text{for } x \in S.$$

- This is the classical probability from Lecture 4: all outcomes equally likely.
- Roulette: b is discrete uniform on $\{0, 1, \ldots, 36\}$, with $p(x) = 1/37$ and $E(b) = 18$.
- A fair die: X is discrete uniform on $\{1, \ldots, 6\}$, with $E(X) = 3.5$ and

$$\mathrm{Var}(X) = \frac{1 + 4 + 9 + 16 + 25 + 36}{6} - 3.5^2 = \frac{91}{6} - 12.25 \approx 2.92.$$

Which Distribution?

Situation	Distribution	$E(X)$	$\mathrm{Var}(X)$
One binary outcome	Bernoulli(p)	p	$p(1-p)$
Successes in n independent trials, same p	Binomial(n, p)	np	$np(1-p)$
Events in an interval, constant rate	Poisson(λ)	λ	λ
Equally likely values in a finite set	Discrete uniform		

- Identify the experiment first. The distribution follows from its structure, not from habit.
- In R, the prefix `d` gives the PMF, `p` gives the CDF, and `r` gives random draws: e.g., `dbinom`, `pbinom`, `rbinom`; `dpois`, `ppois`, `rpois`.

Summary

- A random variable assigns a number to each random outcome. The PMF $p(x) = P(X = x)$ and the CDF $F(x) = P(X \leq x)$ each describe its distribution completely.
- Expected value: $E(X) = \sum_x x\, p(x)$. Linearity: $E(a + bX) = a + b\, E(X)$, but $E[g(X)] \neq g(E(X))$ for nonlinear g.
- Variance: $\mathrm{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$, and $\mathrm{Var}(a + bX) = b^2\, \mathrm{Var}(X)$.
- Two random variables can have the same mean but very different risk.
- Bernoulli for one binary outcome, binomial for the number of successes in n independent trials, Poisson for counts of events at a constant rate, discrete uniform for equally likely values.

Practice Question 1: A Vendor's Profit

A street vendor's daily profit X has the following PMF.

x	-10	20	50
$p(x)$	0.2	0.5	0.3

(a) Verify that this is a valid PMF, and find the CDF $F(x)$ at $x = -10, 20, 50$.

(b) What is the probability that the vendor makes a positive profit?

(c) Find $E(X)$.

(d) Find $\mathrm{Var}(X)$ and the standard deviation of X.

(e) The vendor pays a 20% tax on profit and a fixed stall fee of 5 per day, so the vendor's take-home pay is $Y = 0.8X - 5$. Find $E(Y)$ and the standard deviation of Y.

Practice Question 1: Answers

(a) Each $p(x)$ is between 0 and 1, and $0.2 + 0.5 + 0.3 = 1$. $F(-10) = 0.2$, $F(20) = 0.7$, $F(50) = 1$.

(b) $P(X > 0) = 1 - F(-10) = 0.8$.

(c) $E(X) = -10(0.2) + 20(0.5) + 50(0.3) = -2 + 10 + 15 = 23$.

(d) $E(X^2) = 100(0.2) + 400(0.5) + 2500(0.3) = 970$, so

$$\mathrm{Var}(X) = 970 - 23^2 = 441, \qquad \sigma = 21.$$

(e) $E(Y) = 0.8(23) - 5 = 13.4$, $\quad \mathrm{sd}(Y) = 0.8 \times 21 = 16.8$.

Practice Question 2: Target Practice

A marksman hits a target 80% of the time. The marksman fires 6 shots, and the shots are independent. Let X = the number of shots that hit the target.

(a) What is the distribution of X? State its parameters.

(b) What is the probability that exactly 4 shots hit the target?

(c) What is the probability that more than 4 shots hit the target?

(d) What is the probability that at least 2 shots hit the target?

(e) Find the mean and standard deviation of X. Would it be unusual for none of the shots to hit the target?

Practice Question 2: Answers

(a) $X \sim \text{Binomial}(6, 0.8)$.

(b) $P(X = 4) = \binom{6}{4}(0.8)^4(0.2)^2 = 15 \times 0.4096 \times 0.04 \approx 0.246$.

(c) $P(X > 4) = P(X = 5) + P(X = 6) = 0.393 + 0.262 \approx 0.655$.

(d) Complement rule:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.000064 - 0.001536 = 0.9984.$$

(e) $E(X) = 6 \times 0.8 = 4.8$ and $\sigma = \sqrt{6 \times 0.8 \times 0.2} \approx 0.98$. Zero hits is $(0 - 4.8)/0.98 \approx -4.9$ standard deviations below the mean, and $P(X = 0) = 0.2^6 = 0.000064$. Yes, it would be very unusual.

Practice Question 3: Traffic Accidents

A section of highway has on average 2 traffic accidents per week. Assume that accidents occur independently at a constant rate.

(a) What is the distribution of the number of accidents in one week?

(b) What is the probability of exactly one accident in a week?

(c) What is the probability of no accidents in a week?

(d) What is the probability of at most three accidents in a two-week period?

Practice Question 3: Answers

(a) $X \sim \text{Poisson}(2)$, so $E(X) = \text{Var}(X) = 2$.

(b) $P(X = 1) = e^{-2}\, 2^1/1! = 2e^{-2} \approx 0.271$.

(c) $P(X = 0) = e^{-2} \approx 0.135$.

(d) Over two weeks, the rate is $\lambda = 2 \times 2 = 4$, so $Y \sim \text{Poisson}(4)$:

$$P(Y \leq 3) = e^{-4}\left(1 + 4 + \frac{4^2}{2!} + \frac{4^3}{3!}\right) = e^{-4}(1 + 4 + 8 + 10.67) \approx 0.433.$$